



```
import cv2
from google.colab.patches import cv2_imshow

img = cv2.imread("/content/Screenshot 2026-07-25 083152.png")

sobel_x = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3)

edges = cv2.addWeighted(sobel_x, 0.5, sobel_y, 0.5, 0)

print("Original Image")
cv2_imshow(img)

print("Sobel XY Edge Detection")
cv2_imshow(cv2.convertScaleAbs(edges))

cv2.imwrite("/content/Sobel_XY.jpg", cv2.convertScaleAbs(edges))

print("Image saved successfully")
```

Original Image



Sobel XY Edge Detection

